Lecture 7

Joint Distributions of Discrete Random Variables

23.03.2026

Outline

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Joint cdf

Definition

We have a pair of random variables (either discrete or continuous) X *and* Y*. The joint cumulative probability distribution function of* X *and* Y *is defined by*

$$F_{X,Y}(x, y) = P[X \leq x, Y \leq y]$$

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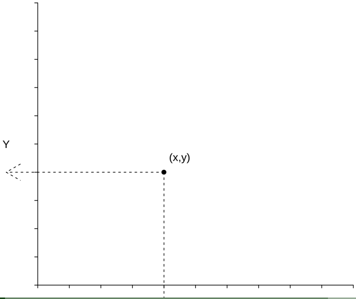
$$\begin{aligned} F_{X,Y}(x,y) &= P[X \leq x, Y \leq y] \\ &= P[(X,Y) \text{ lies south-west of the point } (x,y)] \end{aligned}$$



Properties of joint cdf

- For one random variable: marginal cdf

$$F_X(x) = F_{X,Y}(x, \infty)$$

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- Joint probabilities

$$P(X > x, Y > y) = 1 - F_X(x) - F_Y(y) + F_{X,Y}(x, y)$$

Example: two discrete random variables

Draw two socks at random, without replacement, from a drawer full of twelve colored socks:

6 black, 4 white, 2 purple

Let B be the number of Black socks, W the number of White socks drawn.

Then the pmf of B and W are given by:

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P(B=k)			
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Note - $P(B = k) =$ and $P(W = k) =$

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The joint pmf is given by: $p_{B,W}(b, w) = P(B = b, W = w)$

W

		0	1	2	
	0				
B	1				
	2				

$$P(B = b, W = w) = \begin{cases} & \\ & \\ & \end{cases}$$

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Marginal Distributions

Note that the column and row sums are the distributions of B and W respectively.

$$P(B=b) = P(B=b, W=0) + P(B=b, W=1) + P(B=b, W=2)$$
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$$P(W = w) = P(B = 0, W = w) + P(B = 1, W = w) + P(B = 2, W = w)$$

These are the <u>marginal</u> distributions of B and W. In general,

$$P(X = x) = \sum_y P(X = x, Y = y) = \sum_y P(X = x \mid Y = y) P(Y = y)$$

Conditional Distribution

Conditional distributions are defined as we have seen previously with

$$P(X = x \mid Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{\text{joint pmf}}{\text{marginal pmf}}$$

Question

2. Find the pmf for white socks given no black socks were drawn.

		W			
		0	1	2	
B	0	$\frac{1}{66}$	$\frac{8}{66}$	$\frac{6}{66}$	$\frac{15}{66}$
	1	$\frac{12}{66}$	$\frac{24}{66}$	0	$\frac{36}{66}$
	2	$\frac{15}{66}$	0	0	$\frac{15}{66}$
		$\frac{28}{66}$	$\frac{32}{66}$	$\frac{6}{66}$	$\frac{66}{66}$

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$$P(W = w | B = 0)$$
$$= \frac{P(W = w, B = 0)}{P(B = 0)}$$
$$= \begin{cases} \frac{1}{66} \Big/ \frac{15}{66} = \frac{1}{15} & \text{if } W = 0 \\ \frac{8}{66} \Big/ \frac{15}{66} = \frac{8}{15} & \text{if } W = 1 \\ \frac{6}{66} \Big/ \frac{15}{66} = \frac{6}{15} & \text{if } W = 2 \end{cases}$$

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Independent random variables

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Random variables X *and* Y *are independent if any real sets* $A, B \subset \mathbb{R}$,

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$$

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Independent random variables

The discrete random variables X and Y are independent **if and only if** their joint probability mass function can be expressed as

$$p_{X,Y}(x, y) = p_X(x)p_Y(y),$$

More than two random variables

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